Analysis & Design of Algorithms (Course Code: 3150703)

Uploaded By: Dharmi Patel (5th Sem Computer Science & Engineering) | Official Course Curriculum | Peer Verified & 4.9

Unit 1: Mathematical Foundations & Asymptotic Notations

• Asymptotic Analysis: Big-O (Upper bound), Omega :' „Æðwer bound), and Theta 9, …F–v‡B &÷VæB'à

• Master Theorem for Divide-and-Conquer: T(n) = aT(n/b) + f(n)

- Case 1: If f(n) = O(n^(log_b(a) - ;R''Â F†Vâ T(n) = 9,†åâ†Æöuö"† ''•

- Case 2: If f(n) = 9,†åâ†Æöuö"† '' ¢ Æöuæ²†â''Â F†Vâ T(n) = 9,†åâ†Æöuö"† '' ¢ Æöuâ†²³ '†â'•

- Case 3: If f(n) = :'†åâ†Æöuö"† ' ² µ)) and regularity condition af(n/b) <= cf(n), then T(n) = 9,†b†â'•

• Strassen Matrix Multiplication: 7 multiplications instead of 8, yielding T(n) = 7T(n/2) + O(n^2) = 9,†åã"ãƒ r'à

Unit 2: Dynamic Programming & Solved Problems

• Core Properties: (1) Optimal Substructure, (2) Overlapping Subproblems.

• 0/1 Knapsack Problem: State definition K(i, w) = max(K(i-1, w), val[i] + K(i-1, w - wt[i])). Time Complexity: O(n*W).

• Longest Common Subsequence (LCS): If X[i] == Y[j], dp[i][j] = 1 + dp[i-1][j-1], else max(dp[i-1][j], dp[i][j-1]).

• Matrix Chain Multiplication: Minimizing scalar multiplications for A_1 * A_2 * ... * A_n with O(n^3) DP table.

Unit 3: Greedy Algorithms, Graph Theory & NP-Completeness

• Fractional Knapsack: Greedy choice by value-to-weight ratio (val/wt). Time Complexity: O(n log n).

• Huffman Coding: Optimal prefix code construction using min-priority queue with O(n log n) compression.

• Single Source Shortest Path: Dijkstra Algorithm using Min-Heap: O((V + E) log V). Bellman-Ford for negative edge weights: O(V*E).

• NP-Complete Problems: Satisfiability (SAT), Clique Problem, Vertex Cover, Traveling Salesperson Problem (TSP).